Meraj Nahid Khan

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Education

Iowa State University
Bachelor of Science in Mechanical Engineering
Expected Graduation: Fall 2027 | CGPA: 3.59

Professional Experience

Mechanical Engineering Intern | SystemWorks | West Des Moines, IA
May 2025 – January 2026; May 2026 – August 2026

- Supported mechanical commissioning through design reviews, field inspections, and functional performance testing of HVAC equipment and piping systems.
- Interpreted engineering drawings and specifications to verify mechanical installations, identify deficiencies, and support quality assurance and design compliance.
- Reviewed building automation system (BAS) controls and sequences of operation to verify equipment integration.
- Coordinated technical findings with engineers, clients, and contractors to support cross-functional issue resolution.
- Maintained deficiency logs, tracked corrective actions, and prepared technical reports to document test results and verify mechanical system performance.

Undergraduate Research Assistant | Iowa State University
Bio-AFM and Cellular Mechanics | Jan 2026 – May 2026

- Performed atomic force microscopy (AFM) imaging and force mapping to characterize cell-surface mechanics.
- Quantified surface roughness from AFM images using engineering measurement and experimental data analysis.

Lab Assistant – CCEE | Iowa State University
Sep 2023 – Apr 2025

- Configured monday.com workflows to improve research coordination, task tracking, and laboratory efficiency.
- Automated engineering data reports and laboratory dashboards using Python and VBA for experimental tracking.
- Operated sensor calibration equipment and shop tools to support accurate testing and safe laboratory practices.
- Supported experimental setup, measurement verification, data analysis, and technical reporting.

Undergraduate Teaching Assistant, Thermodynamics II | Iowa State University
Aug 2026 – Present

- Evaluate thermodynamic cycles, energy balances, and variable specific heats; provide technical feedback.

Calculus I Grader | Iowa State University
Jan 2025 – Present

- Assess calculus solutions for mathematical accuracy and analytical reasoning using consistent grading criteria.

Engineering Involvement

Structural Frame Design – PRISUM Solar Car

- Selected 6061-T6 aluminum based on material properties for lightweight, high-strength structural design.
- Designed 3D-printed door-support brackets for additive manufacturing to reduce frame loading and weight.
- Contributed to structural design optimization, balancing component strength, weight, and mechanical integration.

Projects

1/10 Scale RC Car | Personal Design Project | Ongoing

- Design and assemble a 1/10-scale RC car in SolidWorks using 3D CAD and assembly modeling for chassis, steering, suspension, and wheel/axle interfaces.
- Develop custom parts for rapid prototyping and 3D printing using dimensional constraints, assembly mates, and interference checks to verify fit, clearances, and motion.
- Apply design for manufacturing and assembly (DFM/DFA) to refine part geometry and packaging while integrating the motor, servo, wheels, bearings, and hardware.

Flywheel Power Generator | CAD | Iowa State University | Spring 2025

- Designed a flywheel generator in SolidWorks and fabricated a working prototype to convert rotational mechanical energy into electrical power.
- Integrated the frame, flywheel, shafts, bearings, and gears into a mechanical power transmission assembly.
- Soldered a bridge rectifier and capacitors for voltage rectification, energy storage, and electrical integration.

Skills and Interests

Software: SolidWorks, MATLAB, ANSYS, Python, Microsoft Excel (data analysis, charts, VBA), monday.com.
Design and Manufacturing: 3D CAD, assembly modeling, DFM/DFA, 3D printing, rapid prototyping.
Engineering: Thermodynamics, mechanical components, material selection, engineering drawings.
Testing and Analysis: Functional testing, sensor calibration, data analysis, technical documentation.
Interests: Mechanical design, automotive engineering, product development, manufacturing automation.